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# Desigualdad aritmético geométrica por Jensen

**Ejercicio 1.** Sean $\{y_i\}_{i\in\mathbb{N}_n} \subset \mathbb{R}_+$. Demuestra que $\left(\prod_{i=1}^{n} y_i\right)^{1/n} \leqslant \frac{1}{n}\sum_{i=1}^{n} y_i$.

***Solución*:** Tomamos $X = \mathbb{N}_n$, $\Sigma = \mathcal{P}(\mathbb{N}_n)$ y $\mu$ dada por $\forall S \subset \mathbb{N}_n : \mu(S) = {}^{|S|}\!/_{n}$. Entonces, $(X, \Sigma, \mu)$ es un espacio de probabilidad.

Consideramos la función convexa $\varphi(x) = e^x$ y la función $\Sigma$-medible dada por $\forall i \in \mathbb{N}_n$ : $f(i) = \log y_i$. Entonces, por la desigualdad de Jensen, tenemos que

$$\varphi\left(\int_X f\,\mathrm{d}\mu\right) \leqslant \int_X \varphi(f)\,\mathrm{d}\mu \iff \exp\left(\sum_{i=1}^{n} \frac{\log y_i}{n}\right) \leqslant \sum_{i=1}^{n} \frac{e^{\log y_i}}{n}$$

$$\iff \prod_{i=1}^{n} e^{\frac{\log y_i}{n}} \leqslant \frac{1}{n}\sum_{i=1}^{n} y_i \iff \left(\prod_{i=1}^{n} y_i\right)^{1/n} \leqslant \frac{1}{n}\sum_{i=1}^{n} y_i. \qquad \blacksquare$$

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